

Topic 14: Acid-base Equilibria

Knowledge of the concepts introduced in Unit 2, Topic 9B Chemical Equilibria will be assumed and extended in this topic.

Students will be assessed on their ability to:

14.1	understand that a Brønsted–Lowry acid is a proton donor and a Brønsted–Lowry base is a proton acceptor and that acid-base reactions involve proton transfer
14.2	be able to identify Brønsted–Lowry conjugate acid-base pairs
14.3	be able to define the term 'pH'
14.4	be able to calculate pH from hydrogen ion concentration
14.5	be able to calculate the concentration of hydrogen ions in a solution, in mol dm ⁻³ , from its pH, using the expression $[H^+] = 10^{-pH}$
14.6	understand the difference between a strong acid and a weak acid in terms of the degree of dissociation
14.7	be able to calculate the pH of a strong acid
14.8	be able to deduce the expression for the acid dissociation constant, K_a , for a weak acid
14.9	be able to calculate the pH of a weak acid from K_a or pK_a values, making relevant assumptions <i>Students will not be expected to solve quadratic equations.</i>
14.10	be able to define the ionic product of water, K_w
14.11	be able to calculate the pH of a strong base from its concentration, using K_w or pK_w
14.12	be able to define the terms 'pK _a ' and 'pK _w '
14.13	be able to analyse data from the following experiments: i measuring the pH of a variety of substances, including equimolar solutions of strong and weak acids, strong and weak bases, and salts ii comparing the pH of a strong and weak acid after dilution 10, 100 and 1000 times
14.14	be able to calculate K_a for a weak acid from experimental data given the pH of a solution containing a known mass of acid
14.15	be able to draw and interpret titration curves, using all combinations of strong and weak monoprotic and diprotic acids with bases, and apply these principles to diprotic acids and bases
14.16	be able to select a suitable indicator for a titration, using a titration curve and appropriate data
14.17	know what is meant by the term 'buffer solution'
14.18	understand the action of a buffer solution
14.19	be able to calculate the pH of a buffer solution given appropriate data

14.21	understand how to use a weak acid-strong base or strong acid-weak base titration curve to: - i demonstrate buffer action - ii determine K_a from the pH at the point where half the acid is neutralised/ equivalence point
14.22	understand the importance of buffer solutions in biological environments: - i buffers in cells and in blood (H_2CO_3/HCO_3^-) - ii in foods to prevent deterioration due to pH change (caused by bacterial or fungal activity)
14.23	CORE PRACTICAL 11 Finding the K_a value for a weak acid.
	Further suggested practicals: - i carry out the measuring of the pH of solutions mentioned in 14.13 - ii obtain data to draw titration curves mentioned in 14.15 (which gives an opportunity to use data loggers) - iii analysis of vinegar